

A threshold resolution under $\diamond(\mathfrak{b})$

Candidate full answer to Question 1 of arXiv:2504.09351

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Status: full solution likely valid, pending human review. The new ingredient is Lemma 3, a threshold analogue of the source paper’s basic-star guessing condition.

1 Source question and result

Martínez-Celis and Morawski prove under $\diamond(\text{non}(\mathcal{M}))$ that there is a Radon–Nikodým (RN) compact space of weight $\aleph_1$ with a non-RN continuous image [3]. Their construction necessarily has weight at least $\text{non}(\mathcal{M})$, and they ask:

Question 1. Is it consistent to have $\text{non}(\mathcal{M}) > \aleph_1$, but have a Radon–Nikodým compact space of weight $\aleph_1$ with a continuous image that is not Radon–Nikodým?

They specifically suggest changing their condition (5) so that the space can be constructed from $\diamond(\mathfrak{b})$. The relevant passage is reproduced below from page 9 of the source PDF.

This shows that our construction has the smallest weight possible, so if one wants to construct one of these spaces in a similar way, it would be necessary to change the condition (5) from the definition of basic* space.

Recall Avilés’ result from [Avi05], that all quasi Radon-Nikodým spaces are Radon-Nikodým. Our spaces have weight $\text{non}(\mathcal{M})$ and it is known that $\mathfrak{b} \leq \text{non}(\mathcal{M})$ (see [Bla10] for a proof), which leads to the following question.

Question 1. *Is it consistent to have $\text{non}(\mathcal{M}) > \aleph_1$, but have a a Radon-Nikodým compact space of weight $\aleph_1$ with a continuous image that is not Radon-Nikodým?*

One approach to a possible answer to this question would be to, in some way, modify the condition (5), to allow such space to be constructed using $\diamond(\mathfrak{b})$ ($\diamond(\mathfrak{b})$ is the principle $\diamond(\mathbf{B})$, where $\mathbf{B}$ was defined in the introduction).

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We carry out that suggestion.

Theorem 1 (Main theorem). *Assume $\diamond(\mathfrak{b})$. There are compact spaces L_0, L_1 of weight $\aleph_1$ and a continuous surjection $\pi : L_0 \rightarrow L_1$ such that L_0 is Radon–Nikodým and L_1 is not Radon–Nikodým.*

Corollary 2. *It is relatively consistent with ZFC that $\text{non}(\mathcal{M}) > \aleph_1$ and that an RN compactum of weight $\aleph_1$ has a non-RN continuous image. Hence Question 1 has an affirmative consistency answer.*

Indeed, Minami's Corollary 4.7(ii) gives a model in which

$$\mathfrak{c} = \text{non}(\mathcal{M}) = \text{cov}(\mathcal{M}) = \aleph_2 \quad \text{and} \quad \Diamond(\mathfrak{b})$$

hold simultaneously [4, Corollary 4.7(ii), p. 762]. Applying Theorem 1 in that model proves the corollary.

2 Proof intuition

The source construction assigns to each dyadic cell $t \in 2^n$ a continuous Cantor map whose image is the dyadic interval determined by t . Exact guessing was used to arrange that one small cylinder in each relevant fiber is sent to the two endpoints of that interval. Exact strings on a whole level lead naturally to $\text{non}(\mathcal{M})$.

Only the *length* of the small cylinder is actually needed. If S_m deletes the first m bits, then

$$\sigma \mapsto t \cap S_m(\sigma)$$

maps every cylinder of length at most m onto the full Cantor cylinder $[t]$. Thus an integer threshold replaces an exact-string guess.

There are two simultaneous requirements. For each useful level n , one integer $m_\beta(n)$ must work for every $t \in 2^n$; and for every such t we need infinitely many base points, so they converge to the resolution point β . A single $\Diamond(\mathfrak{b})$ coordinate does both jobs. Its value sets the threshold for its own level, while that same successful coordinate adds one new good ladder point to every earlier cell. Infinitely many successful coordinates therefore populate every cell infinitely often.

Crucially, maps belonging to level n still have image diameter at most 2^{-n} . That shrinking-diameter property is exactly what preserves the Reznichenko metric on the Cantor resolution L_0 . On the interval resolution L_1 , lower semicontinuity gives an estimate 4^{-n} across every dyadic cell, and chaining 2^n cells forces the distance between the two endpoints to be at most 2^{-n} on infinitely many levels.

3 The threshold ladder fan

We use the following standard form of $\Diamond(\mathfrak{b})$. For every Borel map $F : 2^{<\omega_1} \rightarrow \omega^\omega$ there is a sequence $\langle u_\beta : \beta < \omega_1 \rangle$, $u_\beta \in \omega^\omega$, such that for every $\gamma \in 2^{\omega_1}$, stationarily many β satisfy

$$F(\gamma \restriction \beta)(j) < u_\beta(j) \quad \text{for infinitely many } j. \tag{1}$$

This is the Borel relational presentation used in [5, 4, 3].

Let $T = 2^{<\omega}$ and, for $t \in T$, put $A_t = \{t\} \times \omega_1$ and $A = \bigcup_{t \in T} A_t$. Let B be the set of nonzero countable limit ordinals. For every $\beta \in B$, fix an increasing cofinal sequence $\langle \lambda_\beta(j) : j \in \omega \rangle$ in β . For $n \leq j$, $t \in 2^n$, fix an enumeration

$$e_{\beta,j,n,t} : \omega \longrightarrow \beta \setminus (\lambda_\beta(j) + 1).$$

Repetitions are harmless, although the enumerations may be chosen bijective.

Lemma 3 (Threshold ladder lemma). *Under $\diamond(\mathbf{b})$ there are countable sets $C_\beta \subset A$ and integers $m_\beta(n)$, for $\beta \in B$ and $n \in \omega$, such that:*

- (i) $C_\beta \cap C_\eta$ is finite whenever $\beta \neq \eta$;
- (ii) $C_\beta \cap A_t$ is infinite for every $t \in T$;
- (iii) for every $\ell : A \rightarrow \omega$ there are $\beta \in B$ and an infinite set $W \subset \omega$ such that, whenever $n \in W$ and $t \in 2^n$,

$$\{a \in C_\beta \cap A_t : \ell(a) \leq m_\beta(n)\} \text{ is infinite.} \quad (*)$$

Proof. We describe the Borel target before applying the diamond. A branch can use a fixed standard code for a countable family $\langle G_t : t \in T \rangle$ of subsets of ω_1 and integers $\langle H(t) : t \in T \rangle$. At a countable stage β , for j, n, t with $n \leq j$ and $t \in 2^n$, let

$$p_{\beta,j,n,t} = \min\{k : e_{\beta,j,n,t}(k) \in G_t \cap \beta\},$$

when this set is nonempty. On correctly formed codes for which all the needed minima exist, set

$$F(\gamma \restriction \beta)(j) = \max\left(\{H(t) : t \in 2^n, n \leq j\} \cup \{p_{\beta,j,n,t} : n \leq j, t \in 2^n\}\right). \quad (1)$$

Set the value equal to 0 on malformed codes. For each fixed β and j , (1) takes a maximum over a finite set, and every displayed minimum is a first-hit search in a fixed enumeration. Hence each coordinate is Borel. The usual pairing of $\omega_1 \times \omega$ with ω_1 makes this one Borel map on $2^{<\omega_1}$.

Let $\langle u_\beta \rangle$ be the $\diamond(\mathbf{b})$ sequence for F , and put

$$m_\beta(n) = u_\beta(n),$$

$$C_\beta[j] = \{(t, e_{\beta,j,n,t}(k)) : n \leq j, t \in 2^n, k \leq u_\beta(j)\}, \quad C_\beta = \bigcup_{j < \omega} C_\beta[j]. \quad (2)$$

Every stage in (2) is finite. If $\beta < \eta$, then C_β has all second coordinates below β , whereas all but finitely many stages of C_η have all second coordinates above β . Thus $C_\beta \cap C_\eta$ is finite, proving (i). For fixed $t \in 2^n$, every stage $j \geq n$ contributes a point of A_t above $\lambda_\beta(j)$; cofinality of the latter sequence gives (ii).

Now fix $\ell : A \rightarrow \omega$. Since A_t has size $\aleph_1$ and $A_t = \bigcup_H \{a : \ell(a) \leq H\}$, choose $H(t)$ so that

$$G_t = \{\xi < \omega_1 : \ell(t, \xi) \leq H(t)\}$$

is uncountable. There is a club of β for which every $G_t \cap \beta$ is cofinal in β and the branch code restricts correctly. Choose in this club a diamond stage β satisfying (1), and let J be the infinite set of successful coordinates j .

If $j \in J$, then $u_\beta(j) > H(t)$ for every t on every level at most j . Moreover, (2) includes, for each such t , a point whose second coordinate is $e_{\beta,j,n,t}(p_{\beta,j,n,t}) \in G_t$. Fix $n \in J$ and $t \in 2^n$. Every sufficiently large $j \in J$ therefore contributes a point of $C_\beta \cap A_t$ with ℓ -value at most $H(t) < u_\beta(n) = m_\beta(n)$. As these points lie above the cofinal lower bounds $\lambda_\beta(j)$, infinitely many are distinct. Taking $W = J$ proves (iii). $\square$

4 The two resolutions

Give $A \cup B$ the locally compact topology in which points of A are isolated and

$$\{\{\beta\} \cup (C_\beta \setminus F) : F \subset C_\beta \text{ finite}\}$$

is a neighborhood base at $\beta \in B$. Lemma 3(i) makes this space Hausdorff. Let $K = A \cup B \cup \{c\}$ be its one-point compactification. This is the source paper's "pre-basic" space, and $w(K) = |K| = \aleph_1$ [3, Lemma 2.3].

Blow up each isolated $a \in A$ to a Cantor fiber:

$$L = (A \times 2^\omega) \cup B \cup \{c\}.$$

The topology has the fiber cylinders $\{a\} \times [s]$ and, for U open in K , the sets $((U \cap A) \times 2^\omega) \cup (U \setminus A)$. Thus L is zero-dimensional compact and has weight $\aleph_1$.

Write $S_m(\sigma)(i) = \sigma(m + i)$ for the tail shift and let ρ be the usual Cantor metric. For $\beta \in B$, define the continuous map $g_\beta : L \setminus \{\beta\} \rightarrow 2^\omega$ by

$$g_\beta(a, \sigma) = t \frown S_{m_\beta(n)}(\sigma) \quad \text{if } a \in C_\beta \cap A_t, |t| = n, \quad (3)$$

and put $g_\beta = \mathbf{0}$ off $(C_\beta \times 2^\omega) \cup \{\beta\}$. Continuity on an isolated fiber is clear. At $\eta \in B \setminus \{\beta\}$, it follows from finiteness of $C_\eta \cap C_\beta$; at c , take a neighborhood avoiding the compact set $(C_\beta \times 2^\omega) \cup \{\beta\}$. Notice the uniform range estimate

$$\text{diam}_\rho g_\beta(\{a\} \times 2^\omega) \leq 2^{-n} \quad (a \in C_\beta \cap A_t, |t| = n). \quad (4)$$

Let $q : 2^\omega \rightarrow [0, 1]$ be $q(x) = \sum_i x(i)2^{-(i+1)}$, and put $h_\beta = q \circ g_\beta$. Define graph resolutions

$$\begin{aligned} L_0 &= \{(u, v) \in L \times (2^\omega)^B : \beta \neq u \Rightarrow v(\beta) = g_\beta(u)\}, \\ L_1 &= \{(u, w) \in L \times [0, 1]^B : \beta \neq u \Rightarrow w(\beta) = h_\beta(u)\}. \end{aligned}$$

They are compact, and applying q coordinatewise gives a continuous surjection $\pi : L_0 \rightarrow L_1$. Both weights are $\aleph_1$.

5 The Cantor resolution is RN

We adapt Proposition 4.1 of Avilés–Koszmider [1] and Theorem 2.6 of the source paper [3]. Define a metric d on L_0 as follows. It is zero on the diagonal; over one $\beta \in B$ use the Cantor distance between the two free β -coordinates; over one isolated base point $a \in A$ use the Cantor distance between the two fiber parameters; and set $d = 1$ in every remaining off-diagonal case.

Exactly as in [1, Proposition 4.1], d fragments L_0 . Indeed, a nonempty subset meeting an isolated Cantor fiber has a relatively open piece of arbitrarily small d -diameter. If it meets no such fiber, project it to the scattered space $B \cup \{c\}$, isolate one projected point, and restrict the corresponding free Cantor coordinate.

It remains to verify the Reznichenko separation property. Distinct points with distinct base points, or two distinct points in one isolated fiber, are handled by disjoint base or Cantor neighborhoods. Consider two distinct points over the same $\beta \in B$, and choose disjoint clopen Cantor neighborhoods of their free β -coordinates. If these lifted neighborhoods had d -distance zero, there would be pairs over common isolated base points $a_i = (t_i, \xi_i) \in C_\beta$ with fiber parameters σ_i, τ_i such that $\rho(\sigma_i, \tau_i) \rightarrow 0$, while $g_\beta(a_i, \sigma_i)$ and $g_\beta(a_i, \tau_i)$ remain in the two separated coordinate neighborhoods.

If $|t_i| \rightarrow \infty$ along a subsequence, (4) makes the two images converge together. Otherwise pass to a subsequence on which $t_i = t$ is fixed. Formula (3) then uses one fixed continuous shift $S_{m_\beta(|t|)}$, so again the two images converge together. Both alternatives contradict separation of the coordinate neighborhoods. Thus d is a fragmenting Reznichenko metric.

It follows that L_0 is quasi-RN. It is zero-dimensional, and every zero-dimensional quasi-RN compactum is RN by Arvanitakis [2, Theorem 3.6]. Hence L_0 is RN.

6 The interval resolution is not RN

Suppose toward a contradiction that an l.s.c. metric δ fragments L_1 . For $a \in A$, write $a + \sigma$ for the unique point of L_1 over $(a, \sigma) \in L$. Fix $n \geq 1$, $t \in 2^n$, and $a \in A_t$. Fragment the Cantor fiber $\{a + \sigma : \sigma \in 2^\omega\}$ at scale 4^{-n} . A nonempty relatively open piece contains a cylinder $[s(a)]$ such that

$$\text{diam}_\delta\{a + \sigma : \sigma \in [s(a)]\} < 4^{-n}. \quad (5)$$

Put $\ell(a) = |s(a)|$ (define it arbitrarily on the level $n = 0$), and apply Lemma 3. Obtain β and an infinite W satisfying (*). Discard 0 from W .

Fix $n \in W$ and $t \in 2^n$. There are infinitely many $a \in C_\beta \cap A_t$ with $|s(a)| \leq m = m_\beta(n)$. For each such a , choose two points of $[s(a)]$ by filling with zeroes up to coordinate m and then using respectively the constant-zero and constant-one tails. Call them x_a^0, x_a^1 . Then

$$S_m(x_a^0) = \mathbf{0}, \quad S_m(x_a^1) = \mathbf{1},$$

so (3) gives

$$h_\beta(a, x_a^0) = q(t \frown \mathbf{0}), \quad h_\beta(a, x_a^1) = q(t \frown \mathbf{1}). \quad (6)$$

Enumerate infinitely many such a without repetition. They converge to β because they lie in C_β . In the graph resolution, continuity of every h_η at β for $\eta \neq \beta$, together with (6), gives

$$a + x_a^0 \longrightarrow \beta \oplus q(t \frown \mathbf{0}), \quad a + x_a^1 \longrightarrow \beta \oplus q(t \frown \mathbf{1}),$$

where $\beta \oplus r$ denotes the point over β with free β -coordinate r . From (5) and lower semicontinuity,

$$\delta(\beta \oplus q(t \frown \mathbf{0}), \beta \oplus q(t \frown \mathbf{1})) \leq 4^{-n}. \quad (7)$$

Order the 2^n words $t \in 2^n$ lexicographically. The right endpoint of one dyadic interval equals the left endpoint of the next: $q(t \frown \mathbf{1}) = q(t^+ \frown \mathbf{0})$. Chaining (7) over all words gives

$$\delta(\beta \oplus 0, \beta \oplus 1) \leq 2^n 4^{-n} = 2^{-n}.$$

This holds for infinitely many, hence unbounded, $n \in W$. Therefore the two distinct points $\beta \oplus 0$ and $\beta \oplus 1$ have δ -distance zero, a contradiction. Thus L_1 is not RN, completing the proof of Theorem 1.

7 Verification notes and limitations

1. The almost-disjointness check is insensitive to the diamond: all points added at stage j lie above the cofinal lower bound $\lambda_\beta(j)$, and each stage is finite.
2. In Lemma 3, one successful coordinate j sets the threshold for level j and simultaneously adds a point from every G_t on all levels at most j . Infinitely many successful coordinates are therefore enough to make every cell on each successful level infinite.
3. The shift is fixed on a given β and dyadic level. This is essential in the bounded-level branch of the Reznichenko argument.
4. No preservation theorem for RN compacta under forcing is used. The spaces are constructed directly inside Minami's $\diamond(\mathfrak{b})$ model.
5. The packet claims a relative-consistency answer, exactly matching the source question. It does not prove in ZFC that such a weight- $\aleph_1$ example exists.

Human-review recommendation. Verify the uniform Borel coding leading to (1), then audit the same- β Reznichenko case and the graph-resolution convergence used before (7). These are the only nonroutine interfaces between the new combinatorial lemma and the published resolution argument.

8 Novelty check

A bounded search was performed in the run’s registry, solution, attempt, and proof-gap indexes; for arXiv:2504.09351 and the exact wording of Question 1; for the core terms `diamond(b)`, `non(M)`, and small RN compact continuous images; and in the source’s citation neighborhood and later arXiv/web records visible through 9 August 2026. No later paper explicitly answering Question 1 was found. Minami [4] supplies the forcing model used in Corollary 2, but does not contain the topological construction proved here.

References

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